

Instructions:

Section C is compulsory. Attempt any six questions selection three questions each from section A & B.

Section A

1. a) Find the octal equivalent of the decimal number (417)₁₀
b) Convert (214)₈ into decimal. [5]
2. What are the features of Higher Level Language over Machine Language? [5]
3. How distributed operating system is similar to network operating system in context of its functionalities? What are other types of operating system? [5]
4. Differentiate global int and static int declaration with example. [5]
5. Write a program for checking whether the given number is palindrome or not without a recursive method. [5]
3x5

Section B

6. Write a program to remove duplicates in an array. [5]
7. Explain polymorphism with its types and examples. [5]
8. What is recursion? Write recursive function that returns the nth Fibonacci number. [5]
9. How to create a class and its object in C++. How private members of the class can be assessed outside the class? [5]
10. Explain the difference between struct and class. [5]
3x5

Section C

11. a) 'If the decimal number is a fraction then its binary equivalent is obtained by dividing the number continuously by 2'. If the statement is incorrect, correct it and give the reason.
b) What is pseudo codes in Assembly Language?
c) Define 'std'?
d) What are references in C++?
e) What is an expression in C?
f) When is void() return type used?
g) Define and list a built-in function in C.
h) Write a program in C to add two numbers without using the addition operator.
i) What is Friend Function?
j) Define encapsulation. 10x2

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Roll No.

Total Pages : 4

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COMPUTER PROGRAMMING

Paper-1103T/1202T

Semester-I

Time Allowed : 3 Hours] [Maximum Marks : 50

Note : The candidates are required to attempt **three** questions each from Sections A and B carrying 5 marks each and the entire Section C consisting of 10 short answer type questions carrying 2 marks each.

SECTION—A

1. What is the difference between Compiler and Interpreter? 5

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2. Name any three Operating systems. Explain them. 5
3. Perform the following conversions : 5
- (a) Convert Octal no. (128.64) into decimal number.
- (b) Convert hexadecimal (2B.C4) into decimal.
- (c) Subtract 35 from 17 by using complimentary method.
4. Write a note on Operators. Explain them with their precedence and associativity. 5
5. Explain all Branching statements with the help of their syntax. 5

SECTION—B

6. What is the difference between C and C++? 5
7. Explain Recursion. W.A.P. to find factorial of a number by using recursion. 5

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8. What is a String? Explain all functions of strings. 5
9. Explain the concept of Classes and Objects with the help of example. 5
10. W.A.P. to find transpose of a matrix. 5

SECTION—C

11. Write short notes on the following : 10×2=20
- (i) What do you mean by Pointers?

- (vii) What do you mean by Text files?
- (viii) Why Storage classes are necessary?
- (ix) Name Access modifiers and explain their working.
- (x) What is an object?